	For 20 m s^{-1} to 30 m s^{-1}	B1
2ai	From the graph, at $4.0\text{x}10^{-8}$m, g = 0.8 N kg^{-1} gravitational force, F = mg = $(5.5\text{x}10^{13})(0.8)$= $4.4 \text{ x } 10^{13}$ N [accept range g = 0.5 N kg^{-1} to 0.9 N kg^{-1} and F between 2.75. $\text{x}10^{13}$ N to $4.95 \text{ x}10^{13}$ N]	A1
2aii	Area under graph from $2.0 \text{ x } 10^{8}$ m to $4.0 \text{ x } 10^{8}$ m = (7.5 squares)$(0.5\text{x}10^{8})(1)$ =$3.75\text{x}10^{8}$ N kg^{-1} m [Accept range 7.5 squares to 8.5 squares, $3.75\text{x}10^{8}$ to $4.25\text{x}10^{8}$ N kg^{-1} m] By Conservation of energy, KE gained=GPE lost = $5.5\text{x}10^{13}$ (area under g-distance graph) = $5.5\text{x}10^{13}$ $(3.75\text{x}10^{8})$ = $2.06 \text{ x}10^{22}$J **Or** use the point values from the graph for calculations.	M1 A1
2bi	Period of rotation of geostationary satellite = 24x3600=$8.64\text{x}10^{4}$ s Gravitational force provides centripetal force for satellite to orbit around earth. $\frac{GMm}{{r_{geo}}^2} = mr_{geo}(\frac{2\pi}{T})^2$ $r_{geo} = [\frac{(6.67\times10^{-11})(6.0\times10^{24})(8.64\times10^{4})^2}{(2\pi)^2}]^{\frac{1}{3}}$ $r_{geo} = 4.23\times10^{7}m \approx 4.23x10^{4}km$	 M1 A0

2bii	Kinetic energy of satellite at the equator = ½ mv^2 = ½ $(1000)((6.39x10^6)\frac{2\pi}{8.64x10^4})^2$ = 1.08×10^8 J Total energy of satellite at the equator = KE + GPE = $1.08 x10^8 + (-\frac{6.67x10^{-11}(1000)(6.0x10^{24})}{6.39x10^6})$ = $-6.2521x10^{10}$ J Note: Total energy of satellite in a geostationary orbit $= -\frac{GMm}{2r} = = -(\frac{1}{2})(6.67\times10^{-11})(6.0\times10^{24})(1000)\left(\frac{1}{4.23\times10^{7}}\right) = -4.7305 \times 10^9$ J Min energy required = total energy of satellite in geostationary orbit – total energy of satellite at the surface of earth = -4.7305×10^9 J $-(-6.2521x10^{10})$ = 5.78×10^{10} J	M1 M1 A1
2biii	E_t becomes more negative radius of orbit *r* will decrease as magnitude of E_t will increase as <u>radius decrease</u>. Since $E_t = -E_k$ and E_t becomes more negative, $E_k = \frac{GMm}{2r}$ will increase and hence <u>speed of satellite v will increase.</u>	M1 A1 A1
3ai 1	Interference pattern of Black and White equally spaced stripes	A1